WORKED SOLUTION

STEP I 2001, Question 1

Pure • Geometric Inequalities

STEP I 2001, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s1-q1

The key idea

Three points on three different sides of a square must use a pair of opposite sides, and the third side joins those two. Put coordinates on the square so that each point has a single free parameter between 0 and 1. Then argue by contradiction: if all three sides of the triangle were longer than the bound, each length condition would squeeze one parameter, and the squeezes together would push a point off the square.

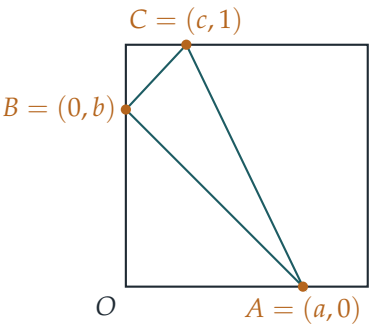
Solution

Work in centimetres. Four sides and three points on different sides means the points miss exactly one side of the square, so two of the points lie on a pair of opposite sides and the third lies on a side adjacent to both. Choose axes with the square as $0 \le x \le 1$, $0 \le y \le 1$, so that

$$A = (a, 0) \text{ on the bottom}, \qquad B = (0, b) \text{ on the left}, \qquad C = (c, 1) \text{ on the top},$$

with $a, b, c \in [0, 1]$. (Rotating or reflecting the square doesn't change any lengths, so this is no loss of generality.) Then

$$AB^2 = a^2 + b^2, \qquad BC^2 = c^2 + (1-b)^2, \qquad CA^2 = (c-a)^2 + 1.$$



Let $k = \sqrt{6} - \sqrt{2}$ and $t = 2 - \sqrt{3}$. Two small facts make the numbers behave:

$$k^2 = 8 - 2\sqrt{12} = 8 - 4\sqrt{3}, \qquad t^2 = 7 - 4\sqrt{3}, \qquad \text{so} \qquad k^2 - 1 = t^2.$$

Also $2t = 4 - 2\sqrt{3} = (\sqrt{3} - 1)^2$, so $\sqrt{2t} = \sqrt{3} - 1$, and therefore

$$\sqrt{2t} + t = (\sqrt{3} - 1) + (2 - \sqrt{3}) = 1.$$

Suppose, for contradiction, that all three sides of the triangle are longer than k. By the symmetry of reflecting the square in the line $y = \frac{1}{2}$ (which swaps the roles of A and C, and of a and c), we may assume $c \geq a$.

From $BC > k$. Since $c \leq 1$,

$$k^2 < BC^2 = c^2 + (1 - b)^2 \leq 1 + (1 - b)^2,$$

so $(1 - b)^2 > k^2 - 1 = t^2$. As $1 - b \geq 0$ and $t > 0$, this gives $1 - b > t$, that is, $0 \leq b < 1 - t$.

From $AB > k$. Since $0 \leq b < 1 - t$ we have $b^2 < (1 - t)^2$, so

$$a^2 > k^2 - b^2 > k^2 - (1 - t)^2 = (1 + t^2) - (1 - 2t + t^2) = 2t.$$

As $a \geq 0$, this gives $a > \sqrt{2t} = \sqrt{3} - 1$.

> **?** **Ask yourself** We've pinned down a; what does the third side CA add?

The side CA crosses the square from bottom to top, so it has vertical extent 1 and must make up the rest of its length horizontally.

From $CA > k$. We have $(c - a)^2 + 1 > k^2$, so $(c - a)^2 > t^2$, and since $c \geq a$ this means $c - a > t$. Hence

$$c > a + t > \sqrt{2t} + t = 1.$$

But C lies on the top side, so $c \leq 1$. This contradiction shows that the three sides can't all be longer than k: at least one side of triangle ABC has length at most $(\sqrt{6} - \sqrt{2})$ cm, as required.

The bound is attained

Every inequality above is tight when $c = 1$ and $a = b = \sqrt{3} - 1$, that is, $A = (\sqrt{3} - 1, 0)$, $B = (0, \sqrt{3} - 1)$, $C = (1, 1)$. Then

$$AB^2 = 2(\sqrt{3} - 1)^2 = 8 - 4\sqrt{3}, \qquad BC^2 = CA^2 = 1 + (2 - \sqrt{3})^2 = 8 - 4\sqrt{3},$$

so ABC is equilateral with side exactly k (with C at a corner, counted as lying on the top side). So $\sqrt{6} - \sqrt{2}$ is the best possible constant. In trigonometric terms, $k = \sec 15^\circ$: this is the largest equilateral triangle that fits in a unit square, tilted at 15° to the sides.

Checks

- $k = \sqrt{6} - \sqrt{2} \approx 1.0353$ is a little more than the side of the square, which is sensible: a triangle with a vertex on each of three sides can have all sides longer than 1, but only just.
- A random search over 2×10^6 triples $(a, b, c) \in [0, 1]^3$ found the shortest side never exceeding 1.0337, below k; the equilateral example above has all three sides equal to 1.03528 to machine precision.
- $\sec 15^\circ = 4/(\sqrt{6} + \sqrt{2}) = \sqrt{6} - \sqrt{2}$, matching the equilateral picture.

Common slips

- Assuming the triangle is equilateral, or that one vertex is at a corner, without proof. The extremal case suggests the answer but doesn't prove the inequality for every configuration.
- Forgetting to justify the set-up: the key structural fact is that three distinct sides must include an opposite pair.
- Taking square roots of $(1 - b)^2 > t^2$ or $(c - a)^2 > t^2$ without saying which sign is involved.

Beyond STEP

This is a small *max–min* problem: we are maximising the shortest side over a compact set of configurations, and the extremal configuration makes all the constraints equal at once, a pattern that runs through optimisation and game theory. Because $[0, 1]^3$ is closed and bounded and the shortest side is a continuous function, a maximum must exist, a fact proved in a first analysis course (the extreme value theorem). Abbott's *Understanding Analysis* covers this well, and Siklos's *Advanced Problems in Mathematics* has more STEP geometry in the same spirit.

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